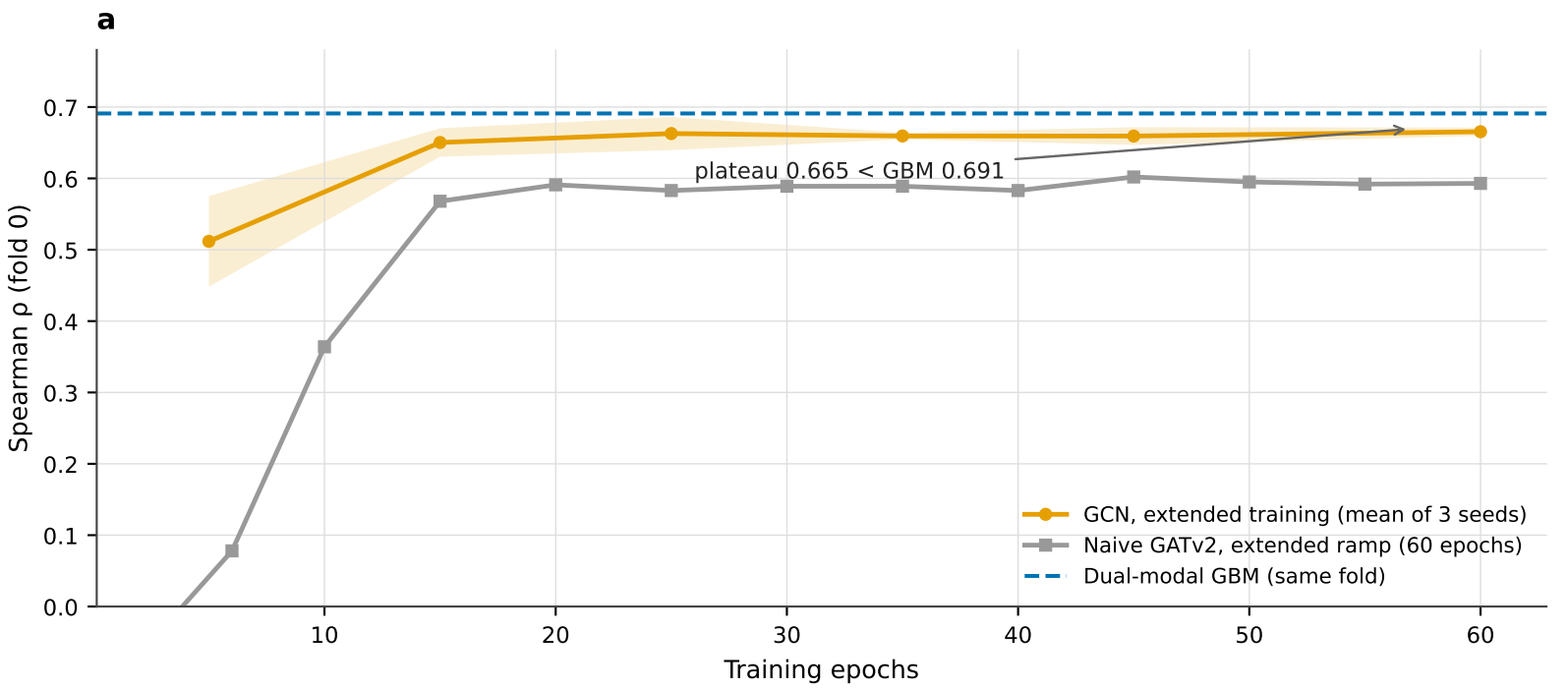
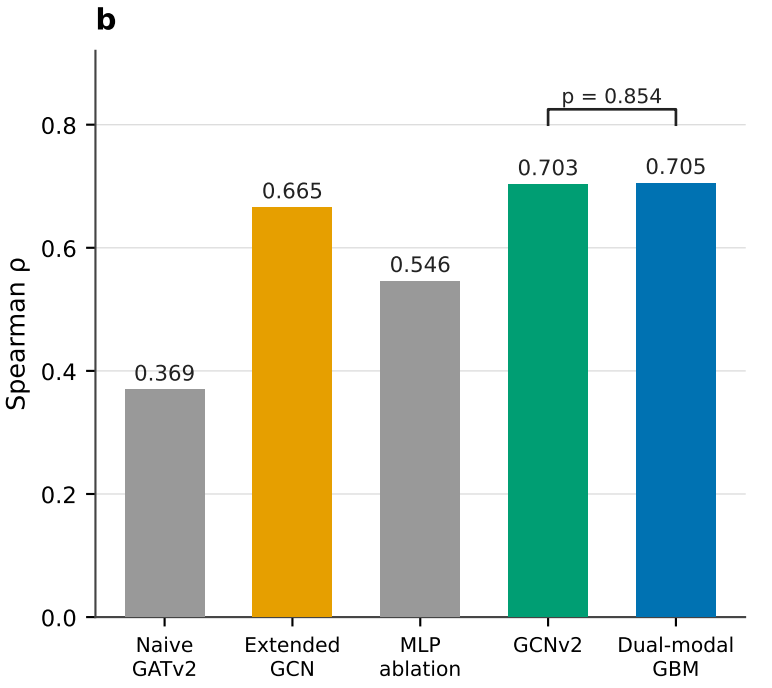
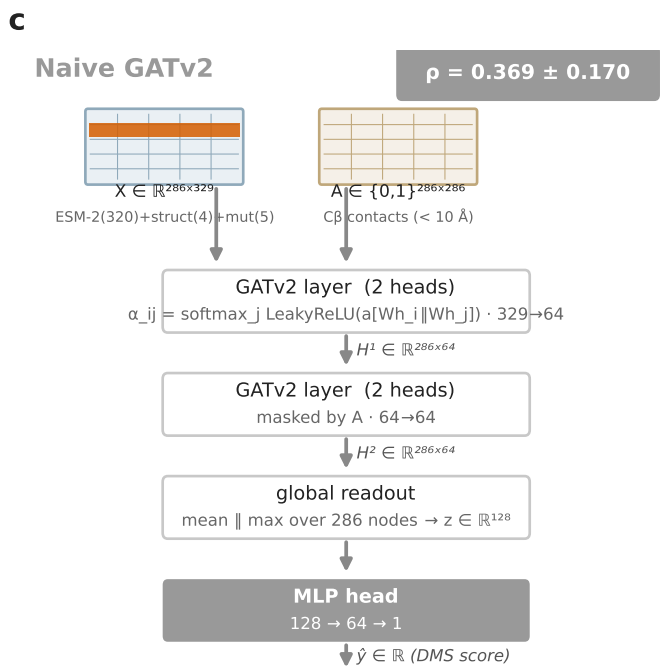
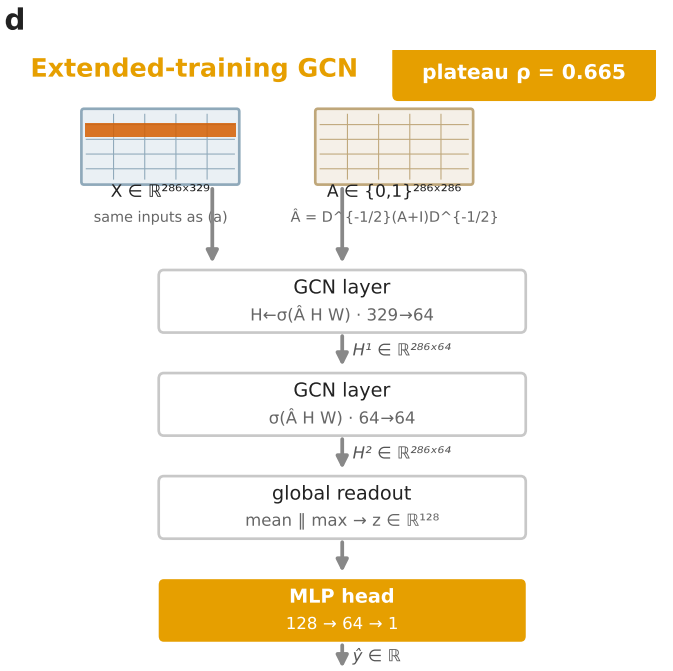
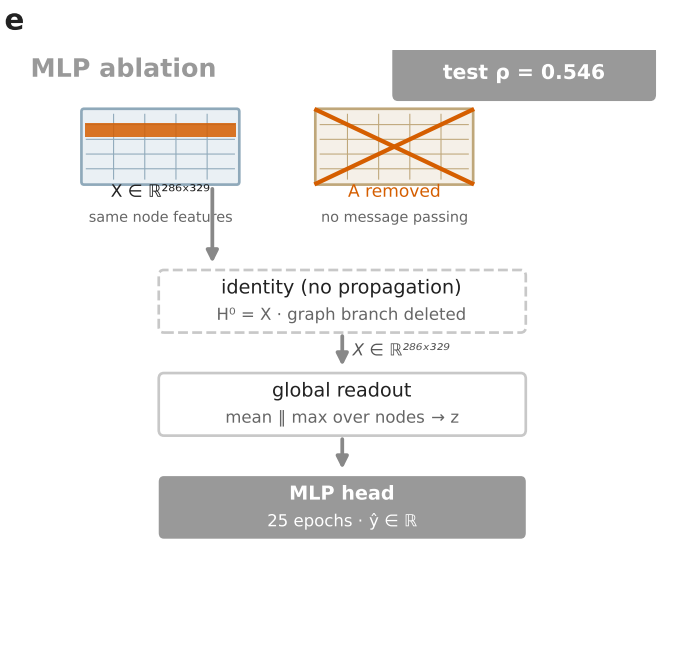






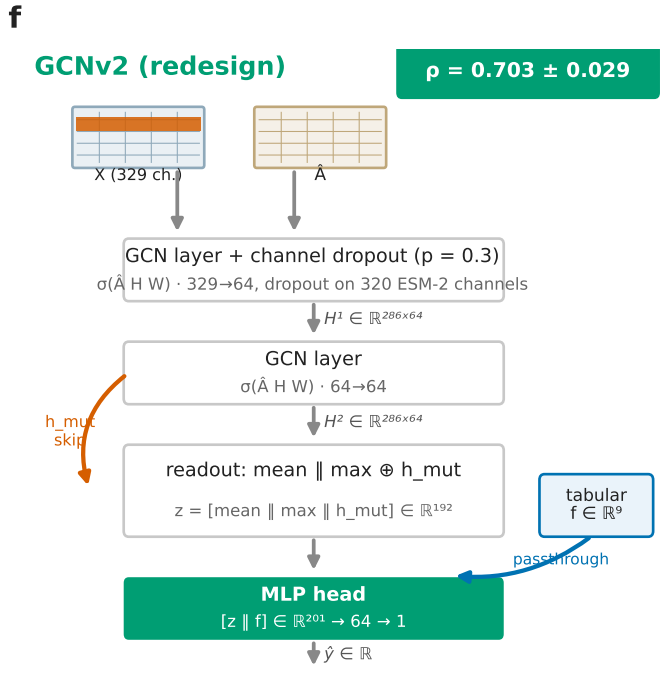


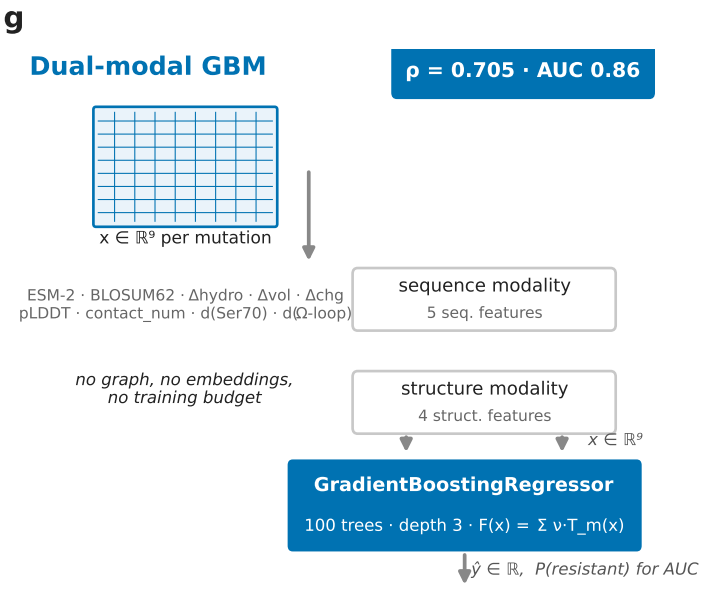


Diagnosis	mutation signal diluted over ~18 neighbours in readout
Diagnosis	position identity memorised from embeddings
Diagnosis	re-learns the pre-digested ESM-2 score from scratch

Finding	architecture unchanged — only the budget grew:
Protocol	60 epochs, lr decay, full 4,000-sample training set
Finding	0.512 (5 ep) → 0.665 plateau, still < GBM 0.691

Diagnosis	train 0.902 vs test 0.546 — gap 0.36
Diagnosis	message passing itself aids generalisation;
Diagnosis	position memorization from embeddings confirmed





Fix	skip beats dilution · passthrough ends re-learning
Fix	channel dropout blocks position memorization

Takeaway	interpretable, trains in seconds on CPU
Takeaway	statistical parity with GCNv2 (paired t-test p = 0.854)
Takeaway	the practical default; GCNv2 the validated graph alternative

h

How to read this figure

Results

a–b: what happened — training dynamics and the final ranking on TEM-1.

Diagnosis

c, e: why the naive end-to-end GNN failed — dilution, memorization, re-learning.

Diagnosis

d: more compute alone does not fix it (plateau 0.665 < GBM 0.691).

Fix

f: three targeted changes restore the graph model to 0.703 ± 0.029.

Validation

g: parity with the interpretable GBM (p = 0.854) closes the loop.

Colours match across all panels: grey = naive/ablation, orange = extended training, green = GCNv2, blue = GBM.